



CHIROPRACTIC
SLEEP ACADEMY

CHIROPRACTIC SLEEP ACADEMY®

How Oral Appliances Interact With the Whole Skeleton

A Biomechanical Primer for Sleep Appliance Manufacturers

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PURPOSE

Why This Workbook Exists

Your appliance does its job at the airway. But it doesn't land on an airway — it lands on a **whole skeleton**.

When that skeleton fights the device, you see it as intolerance, remakes, TMJ complaints, and "it didn't help." Most of those aren't design flaws — they're **biomechanical mismatches** no one measured. This primer helps you:

- reduce device failure
- reduce clinician complaints
- improve outcomes
- differentiate your brand
- support interdisciplinary care

You're not being asked to change what your device does. You're being given the **missing lens** on the body it acts through.

SECTION 1

Why Airway Success $\neq$ Biomechanical Success

An appliance can open the airway on a bench, in a CBCT, or on night-one titration — and still fail in the body over weeks. Airway patency is a **local** win. Whether the patient's **whole structure** accepts and holds that new jaw position is a **global** question. A device that wins locally but loses globally shows up in your return and remake numbers.

| *Success has two scores. Most programs only keep one.*

SECTION 2

How Mandibular Advancement Affects Cervical Load

Advancing the mandible changes the resting pull on the hyoid, the suprahyoid and infrahyoid muscles, and — through that chain — **cervical posture and load**. A jaw held forward night after night can shift head-neck mechanics. In a body already carrying forward-head posture or cervical hypolordosis, that added load is where morning neck pain, headaches, and "I can't wear it" come from. The device didn't fail; the **neck it loaded** was never assessed.

SECTION 3

Cranial Motion and Nasal Resistance

Nasal resistance drives mouth breathing, and mouth breathing undermines appliance outcomes. Cranial and maxillary mechanics influence the nasal airway. When nasal resistance isn't addressed, a mandibular device is compensating for a problem upstream of it — a common reason a well-made appliance underperforms. *(Cranial-motion effects are CSA's model under investigation — a lens to evaluate, not a settled claim.)*

SECTION 4

Pelvic Compensation After Appliance Use

The body is a connected tension system. A change at the jaw can be **compensated for far away** — including at the pelvis and sacroiliac joints — because cranial and pelvic mechanics are linked through the spine and its membranes. When a patient "just doesn't feel right" on an otherwise good device, a downstream compensation is often why. It's invisible in the mouth and on a CBCT.

SECTION 5

Posture Changes Dentists Never See

Dentists work in the oral cavity; they rarely photograph standing posture before and after an appliance. Yet posture is where global adaptation shows up first — head carriage, shoulder level, pelvic rotation. A simple pre/post postural record (a Doctor of Chiropractic's routine finding) turns "the patient complained" into **data your R&D can use.**

SECTION 6

When Appliance Choice Matters Biomechanically

Not every patient should get every device. Dental tolerance (occlusion, perio, restorations, TMJ) is the dentist's call; **biomechanical tolerance** (cervical load capacity, cranial/pelvic response) is the Doctor of Chiropractic's. The best device selection is a **co-decision** informed by both. Your product line performs better when it's matched to the right biomechanical candidate — and worse when it isn't.

SECTION 7

Feedback to Request From Doctors of Chiropractic

The Doctor of Chiropractic sees what your clinicians and your telemetry can't: the **structural response** over weeks. Ask for it. Useful, de-identified feedback:

- pre/post **posture** and cervical-curve notes,
- **TMJ / cervical load** response to advancement,
- signs of **downstream compensation** (pelvic, cranial),
- correlation with **physician-read HST** change (AHI/ODI/SpO₂),
- **responder vs. non-responder** patterns by device type.

This is a feedback loop your competitors don't have.

SECTION 8

Designing Devices With Biomechanical Awareness

Carry the lens into design: titration ranges that respect cervical load; guidance that pairs your device with nasal-airway and postural screening; instructions that invite co-management rather than assume the mouth is the whole story. Devices designed with the whole skeleton in mind **fail less and get recommended more.**

WORKING TOGETHER

The Clean Model

CSA offers a structured way to add this lens — framed around **risk reduction and outcome improvement**, not influence:

- **CSA Manufacturer Education License** (annual) — this workbook, training sessions, and a structured feedback framework.
- **CSA Training for your Clinical Trainers** — so your field team can speak the biomechanical language to dentists.
- **A biomechanical-compatibility review pathway** — an independent, criteria-based evaluation.

What you're actually buying: not influence — risk reduction, better outcomes, and brand differentiation, in a category where everyone else is only measuring the airway.

Note. Any CSA endorsement or compatibility badge is attorney-designed, with clear criteria, disclaimers, and revocation terms. Sleep studies remain physician-read; nothing here is a cure claim; devices are honored — the goal is to make them work better.